Topic Modeling of French Legislative Campaign Manifestos (1981–1993): A Comparative Study of LDA, NMF, and BERTopic

Antoine GILSON
ENSAE Paris
antoine.gilson@ensae.fr

Abstract

This project applies unsupervised topic modeling to a corpus of approximately 8 500 French legislative campaign manifestos (*professions de foi*) from the elections of 1981, 1988, and 1993, drawn from the Archelec archive (Sciences Po / CEVIPOF). We compare three complementary approaches : Latent Dirichlet Allocation (LDA), Non-negative Matrix Factorisation (NMF), and BERTopic, and evaluate them using coherence metrics (c_v, u_{mass}). NMF achieves the highest topic coherence ($c_v = 0.70$) and produces the most interpretable thematic structure. We then cross-reference the discovered topics with candidate metadata (political family, election year, gender, profession), revealing patterns of partisan specialisation and thematic evolution across a pivotal decade of French politics.

1 Introduction and Problem Definition

Campaign manifestos (*professions de foi*) are official documents distributed to every voter before French legislative elections. They constitute a unique window into the programmatic discourse of candidates across the political spectrum: unlike parliamentary debates or press interviews, they are systematically collected, uniformly structured, and exhaustively archived. The **Archelec** project (Sciences Po / CEVIPOF) has digitised and transcribed these documents using Optical Character Recognition (OCR) for the entire Fifth Republic, yielding a large-scale textual corpus with rich metadata.

For this work, we focus on three elections : 1981, 1988, and 1993. This era spans the Mitterrand era and capture major political shifts: the arrival of the Left in power (1981), the first cohabitation and its aftermath (1988), and the landslide return of the Right (1993). During this period, new political forces emerged (the ecologist movement, the rise of the Front National), and the dominant themes of French public debate evolved considerably (from unemployment and nationalisation to immigration and European integration).

Research question. Given this corpus of $\sim$12 500 documents we ask: *can unsupervised topic modeling techniques extract coherent and interpretable political themes from OCR-transcribed campaign manifestos, and do these themes correlate with partisan affiliation, temporal context, and candidate characteristics?*

This task obviously requires NLP techniques because the corpus is too large for manual content analysis, the documents are noisy (OCR artifacts), and we seek to discover latent thematic structures rather than impose predefined categories. The goal is potentially to unveil unknown correlations or evolution from political families.

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

2 Data: Retrieval, Formatting, and Statistical Description

2.1 Data sources

The data come from two complementary sources:

- **Metadata CSV** (33 031 rows, 42 columns): downloaded from `archelec.sciencespo.fr/explorer`. Each row corresponds to one manifesto and includes fields such as election date, department, constituency, candidate name, sex, age, profession, current mandates, and political affiliation (`titulaire-soutien`).
- **OCR transcriptions**: plain-text files produced by OCR on the scanned originals, available via the GitLab repository `arkindex_archelec`. Each file is identified by a unique document ID that matches the CSV.

We restrict the analysis to the three elections for which both metadata and transcriptions are jointly available: 1981, 1988, and 1993. After joining the two sources on the common document identifier, we obtain 12 497 documents with both text and metadata (only 24 CSV entries lack a corresponding text file, and 249 text files lack a CSV match).

2.2 Preprocessing

OCR cleaning. The raw transcriptions contain systematic artifacts introduced by the digitisation process: repeated headers (`Sciences Po / fonds CEVIPOF`), legal boilerplate (*"Ce document n'est pas un bulletin de vote"*), hyphenated words split across lines (`indus- triels` → `industriels`), isolated page numbers, and generic election headers. We remove these patterns with targeted regular expressions. After cleaning, the mean document length decreases from 635 to 617 words, confirming that the removed content was indeed boilerplate rather than substantive text.

Language filtering. A small number of documents from Alsacian constituencies are written in German. Since our NLP pipeline targets French, we apply automatic language detection (`langdetect`) on the first 500 characters of each document and discard non-French texts (111 documents, $<1\%$ of the corpus).

Political family mapping. The `titulaire-soutien` field encodes the candidate's declared political affiliation, but uses free-text party names with high variability (over 200 distinct values). A key part of the job is to exhaustively construct manuallt a dictionary mapping each party to one of five political families: **Left** (PS, PCF, MRG…), **Right** (RPR, UDF, PR…), **Far left** (LCR, LO, PT…), **Far right** (FN, PFN…), and **Greens** (Les Verts, Génération Écologie…). Documents whose affiliation is missing or unrecognised are unfortunately unusable. Indeed, their labelling would require almost a whole NLP project where we would assign the document on its most likely political party. However, this is quite contrary to our analysis where we try to unbias political families at first. These articles are thus labelled *Other / Unknown* (31% of the corpus, mostly candidates who declared "non mentionné"). These are excluded from the following cross-analyses, yielding a working corpus of 8 546 documents.

Tokenisation and lemmatisation. Following the preprocessing pipeline demonstrated in the course (Lemmatization + Stop Word + CountVectorizer), we process the cleaned texts with spaCy (`fr_core_news_md`), retaining only alphabetic tokens of at least 3 characters, excluding standard French stopwords (NLTK + spaCy defaults) and a curated list of corpus-specific filler words (*candidat, élection, vote, député,* month names, etc.) that appear in nearly every manifesto without carrying discriminative information. Each token is replaced by its lemma. As shown in the course, removing stopwords and applying lemmatisation dramatically improves topic quality.

2.3 Descriptive statistics

Table 1 summarises the corpus composition.

Several patterns are notable. First, the total number of documents increases markedly from 1981 to 1993, reflecting the growing number of candidates (especially from the Greens and the Front

Table 1: Corpus composition by election year and political family.

	Left	**Right**	**Far left**	**Far right**	**Greens**
1981	1 076	722	279	68	72
1988	974	690	299	323	199
1993	1 054	892	344	864	690
Total	3 104	2 304	922	1 255	961

National). Second, the Left and Right families dominate the corpus, consistent with the bipolar structure of French legislative elections. Third, the Far right grows from a marginal presence in 1981 (68 documents) to a substantial share in 1993 (864), mirroring the electoral breakthrough of the Front National.

Document lengths are approximately normally distributed (mean: 265 tokens, std: 124), with a long right tail corresponding to multi-page manifestos. The distributions are similar across political families, though Far-left documents tend to be slightly longer and Green documents slightly shorter.

3 Related Work - State of the Art

3.1 Topic modeling: from count-based to neural methods

Topic modeling refers to a family of unsupervised methods that discover latent thematic structures in document collections. The field has evolved through several generations, as surveyed by Churchill and Singh Churchill and Singh [2022]: Latent Semantic Indexing (LSI, Deerwester et al., 1990) introduced the idea of reducing a term-document matrix to a low-dimensional space, but allowed negative values that complicated interpretation. Probabilistic Latent Semantic Indexing (pLSI, Hofmann 1999) added a generative story, which Blei, Ng, and Jordan Blei et al. [2003] extended with Dirichlet priors to produce LDA that is still the most widely used topic model two decades later. Non-negative Matrix Factorisation (NMF, Lee and Seung Lee and Seung [1999]) offered a non-probabilistic alternative that factorises a TF-IDF matrix into sparse, parts-based components. More recently, the availability of pre-trained transformer embeddings enabled clustering-based approaches such as Top2Vec (Angelov, 2020) and BERTopic Grootendorst [2022], which bypass bag-of-words representations entirely.

Other directions exist but were not pursued in this project. Structural Topic Models (STM, Roberts et al. 2014) incorporate document-level covariates directly into the generative process, which would be natural for our metadata-rich corpus but falls outside the course curriculum. Dynamic Topic Models (DTM, Blei and Lafferty 2006) model topic evolution as a continuous process over time; with only three discrete elections, our corpus is better suited to post-hoc temporal cross-tabulation than to continuous dynamics. Guided or seeded LDA (Jagarlamudi et al. 2012) allows injecting prior knowledge through seed words, which could improve topic interpretability on political texts but introduces subjectivity in seed selection. Finally, neural topic models such as ProdLDA (Srivastava and Sutton 2017) replace the Dirichlet prior with a variational autoencoder, but empirical comparisons on short documents have not shown consistent gains over NMF.

3.2 Evaluating topic quality

Topic modeling is unsupervised, so topic quality cannot be measured against ground-truth labels. Fitting metrics such as perplexity (LDA) or reconstruction error (NMF) measure how well the model explains held-out data, but Röder, Both, and Hinneburg Röder et al. [2015] showed that these do not correlate with human judgments of topic coherence. Their systematic study at WSDM 2015 explored over 200 combinations of coherence measures and found that C_v best predicts human ratings. C_{UMass}, based on document co-occurrence counts, provides a complementary internal measure. We use C_v as our primary metric and C_{UMass} as a secondary one throughout this project, computed via gensim's `CoherenceModel`.

3.3 Topic modeling on political texts

Computational analysis of political texts has grown into a subfield of its own. Grimmer and Stewart Grimmer and Stewart [2013] provide the reference survey, emphasising that unsupervised methods are valuable for exploratory analysis of large political archives but require careful human validation : a principle we follow through manual topic labelling and metadata cross-checks. Quinn et al. Quinn et al. [2010] applied dynamic topic models to U.S. Senate speeches, demonstrating that topic proportions track legislative cycles. Catalinac Catalinac [2016] applied LDA to Japanese election manifestos and showed that topic shifts correlate with electoral reform, establishing a precedent for studying campaign discourse through topic modeling.

In a European setting, Greene and Cross Greene and Cross [2017] compared LDA and NMF on European Parliament debates and found that NMF's deterministic nature and sparsity constraints produce more stable and interpretable topics, particularly on short and focused documents. This finding is directly relevant to our corpus of campaign manifestos, which average 240 tokens. BERTopic, while powerful for semantic clustering, tends to produce a large outlier class on corpora with high lexical variability, as we observe in our experiments.

In the French context, the Archelec project (Gaultier-Voituriez, Sciences Po / CEVIPOF) has made digitised electoral archives of the Fifth Republic available for research, but systematic topic modeling with cross-referencing against candidate metadata (party, gender, profession) remains largely unexplored. This gap motivates the present study.

4 Proposed Models

We compare the three topic modeling paradigms presented in the course, each representing a distinct approach to the problem. The choice is motivated both by the course curriculum and by the literature: LDA as the standard probabilistic baseline, NMF as the method most suited to short political documents, and BERTopic as the current embedding-based approach. Table 2 summarises their differences.

4.1 LDA

Latent Dirichlet Allocation Blei et al. [2003] models each document as a mixture θ_d over K topics drawn from a Dirichlet prior (α), where each topic k is a distribution β_k over the vocabulary drawn from a second Dirichlet prior (η). For each word position, a topic is sampled from θ_d and a word from β_k. The bag-of-words assumption discards word order; inference recovers θ and β via variational methods or Gibbs sampling Blei [2012].

We use gensim's `LdaMulticore` rather than scikit-learn because it provides native access to the `CoherenceModel` API. The input is a BoW corpus built from lemmatised tokens via gensim's `Dictionary`, filtered to remove terms in fewer than 5 documents or more than 60% of documents (18 387 terms retained). We train models for $K \in \{5, 7, 9, \ldots, 15\}$ and select the K maximising C_v.

LDA serves as our baseline. Its probabilistic topic mixtures per document are useful for soft assignments, but the literature and our results confirm that it produces more diffuse topics than NMF on this type of corpus.

4.2 BERTopic

BERTopic Grootendorst [2022] bypasses bag-of-words representations through a four-step pipeline: (1) embed documents using a pre-trained sentence-transformer. We use `paraphrase-multilingual-MiniLM-L12-v2`, a lightweight multilingual model; (2) reduce dimensionality with UMAP McInnes et al. [2018] (to 5 components, cosine metric, `min_dist`=0.0); (3) cluster with HDBSCAN Campello et al. [2013] (`min_cluster_size`=10), which determines the number of clusters automatically and assigns outliers to topic -1; (4) extract topic labels via class-based TF-IDF (c-TF-IDF) using a French `CountVectorizer` with the same stopword list as LDA and NMF.

A key difference is that BERTopic operates on raw cleaned text, not lemmatised tokens, since transformer embeddings capture morphology and context internally. The `CountVectorizer` is only used for the final labelling step.

4.3 NMF

Non-negative Matrix Factorisation Lee and Seung [1999] decomposes the document-term matrix $V \in \mathbb{R}_{\geq 0}^{n \times m}$ into a dictionary matrix W (words $\times$ topics) and an activation matrix H (topics $\times$ documents), minimising reconstruction error under non-negativity constraints: $V \approx WH$. Unlike LDA, NMF makes no probabilistic assumptions and operates on TF-IDF rather than raw counts, which downweights common terms before factorisation.

Following the course pipeline, the input is a TF-IDF matrix capped at 10 000 features (`min_df`=5, `max_df`=0.6, unigrams and bigrams). Based on Greene and Cross's findings on European Parliament data and the short average document length in our corpus (240 tokens), we expect NMF to outperform LDA in topic sharpness.

Table 2: Comparison of the three topic modeling approaches (adapted from the course).

Aspect	**LDA**	**NMF**	**BERTopic**
Model type	Probabilistic generative	Matrix factorisation	Embedding-based clustering
Objective	Maximise log-likelihood	Minimise reconstruction error	Density-based clustering + c-TF-IDF
Assumptions	Multinomial + Dirichlet	Non-negativity only	Semantic embeddings + density
Input	BoW (counts)	TF-IDF	SBERT embeddings
K selection	Fixed, grid search on C_v	Fixed, grid search on C_v	Automatic (HDBSCAN)
Topic sparsity	Diffuse	Sharp, localised	Sharp if dense clusters; large outlier class

5 Experiments - Implementation of the proposed models

5.1 Hyperparameter selection

As discussed previously, the number of topics K is a meta-parameter that must be optimised, and both the number of topics and the number of topic words should not be too large for human interpretation.

LDA. The c_v coherence grid search over $K \in \{5, 7, \ldots, 15\}$ yields the best score for K=7. Yet, the curve doesn't exhibit a large advantage for one K compared to another, with a slight peak at $K_{LDA} = 7$. We select K=7 for LDA.

BERTopic. BERTopic does not require a predefined K; the number of topics is determined automatically by HDBSCAN, which builds a cluster hierarchy and selects the optimal number of clusters. With the chosen hyperparameters, the model discovers 149 micro-topics, with 36% of documents assigned to the outlier cluster (-1). As noted in the course, HDBSCAN drops points from clusters as outliers, which is consistent with this observation.

NMF. We perform a similar grid search, evaluating c_v, u_{mass}, and c_{npmi}. However, computing the c_v score was too computationally expensive for NMF. In contrast, u_{mass} is computed directly from document co-occurrence counts and runs in seconds. As a result, we select the K that **maximises** the u_{mass} score (least negative, since $u_{mass} \leq 0$). Here $K_{NMF} = 13$. This is a satisfying result as K_{NMF} is close to K_{LDA}, so we will be able to compare and to interpret results from close perspectives in terms of topics.

5.2 Quantitative evaluation

Table 3 reports the coherence scores for all three models.

NMF achieves the highest c_v score (0.665), confirming that NMF with TF-IDF excels on short, focused political documents. That is consistent with the course observation that NMF produces

Table 3: Coherence scores for the three topic models.

Model	c_v	u_{mass}
LDA (K=7)	0.47	-0.9420
NMF (K=13)	**0.665**	-1.88
BERTopic (149 topics)	0.555	-6.222

"sharper, more localized topics." LDA performs reasonably ($c_v = 0.47$) but produces less crisp topics, which is expected given LDA's "often diffuse topics" characterisation in the course comparison table. BERTopic's low c_v (0.555) is partly attributable to its very fine granularity (149 topics): the per-topic word lists are coherent within each micro-cluster, but the aggregated score over many small topics is penalised. LDA has the best u_{mass} score (-0.9420 vs -1.88 for NMF), which is expected since u_{mass} is computed directly from the BoW co-occurrence statistics that LDA optimises.

5.3 Qualitative comparison of topics

LDA topics (K=7). The LDA model produces 7 topics, several of which are interpretable: an immigration and security topic (dominated by *immigration, immigré, impôt, insécurité*), a workers' struggle and employer critique topic (*travailleur, patron, patronat, salaire, payer*), and an ecology and political alternation topic (*écologie, mitterrand, ouvrier, environnement*). However, two topics (2 and 3) conflate local governance terms with generic political vocabulary (*social, emploi, pays, conseiller*), acting as catch-all residual clusters rather than coherent thematic pillars. The far-right lexical field also splits across two topics rather than consolidating cleanly. Despite this semantic overlap and some low-interpretability clusters, the LDA baseline at $K = 7$ performs reasonably and surfaces politically meaningful axes.

BERTopic. BERTopic discovers 149 micro-topics, many of which correspond to individual party templates identical manifestos distributed by national parties (FN, LO, Les Verts) to all their candidates. This "national template" phenomenon is an interesting finding in itself: BERTopic's semantic embeddings detect near-duplicate documents that LDA and NMF absorb into broader topics. However, the high outlier rate (36%) and extreme granularity make BERTopic less suited for downstream cross-analyses.

NMF topics (K=13). NMF produces more sharply delineated topics, consistent with the course's characterisation. Among the 13 topics, we identify: electoral logistics (*maire, union, conseiller, dimanche*), an animal-rights/ecologist niche (*animaux, nature, rassemblement nature*), a Front National identity topic (*front, nationale, convictions*), an immigration and fiscal topic (*immigration, immigrés, impôts*), and several socioeconomic topics covering employment, public services, and local development. Also, the sanity checks are passed as the NMF detects the green and the far right topics well. However, despite improved interpretability and semantic precision, residual noise and outliers remain and the modeling is far from being perfect.

Selected model for cross-analyses. Given its slight superior coherence score, sharper topic boundaries, and manageable number of topics, we select NMF (K=13) as the primary model for the metadata cross-analyses. Moreover, we might also keep the LDA to limit the variance in the results that might be induced by following only one model's observations.

6 Discussion of the results

Using the topic distributions extracted from the NMF and LDA models, each document is assigned its dominant topic. Cross-tabulating these assignments with candidate metadata or political party enables the quantification of thematic propensity and partisan specialization, effectively mapping the gap between general political discourse and specific group focal points.

6.1 Thematic Propensity by Party

We compute $P(\text{topic} \mid \text{family})$ via normalised cross-tabulation for each of the five political families.

Single-issue parties show extreme thematic concentration. The Far right dedicates 91% of its discourse to just two topics : Immigration/Security (0.47) and National Identity (0.44). This is consistent with the Front National's campaigning strategy. The Greens are equally concentrated on the opposite end: Ecological Transition (0.54) and Animal Welfare (0.43) account for 97% of their discourse, reflecting the unified national platform of the Entente des Écologistes coalition.

The Far left is thematically dispersed but ideologically coherent. Rather than concentrating on one or two topics, it spreads across Social Conflicts (0.25), Public Services (0.24), Workers' Struggles (0.17), Political Alternation (0.15), and Economic Sovereignty (0.10). This five-topic profile reflects the Far left's broader ideological ambition : addressing class struggle, public sector defence, and anti-capitalist critique simultaneously. This contrasts to the Far right's narrower agenda.

Governing parties are thematically broad. The Left distributes across Political Alternation (0.39), Public Finances (0.17), and Democratic Renewal (0.15). This portfolio reflects both national policy discourse (Mitterrand's legacy, public services) and local constituency campaigning. The Right's profile is dominated by the residual Local Elections topic (0.93), which largely captures generic institutional language (*conseiller général*, *maire*, *suppléant*) rather than a substantive thematic choice; excluding this catch-all category, the Right's thematic footprint is too thin to draw strong conclusions from NMF alone. This a major flaw from our model here.

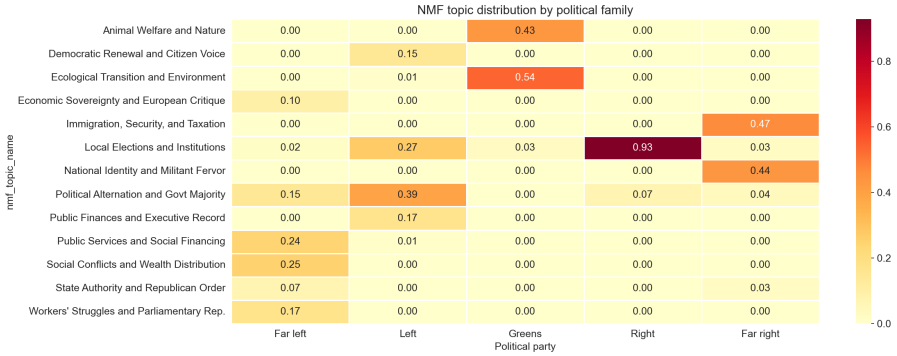


Figure 1: NMF topic distribution by political family. Each row shows $P(\text{topic} \mid \text{family})$.

6.2 Temporal Evolution (1981 → 1988 → 1993)

Tracking NMF topic proportions across the three elections reveals a political landscape in transformation (Figure 2).

Rise of identity and security themes. The Immigration/Security and National Identity topics are virtually absent in 1981, together accounting for less than 5% of the corpus. By 1988 they dominate over 50% of the thematic space, and remain at approximately 15% in 1993 alongside new emerging themes. This trajectory directly mirrors the electoral rise of the Front National, which went from a marginal force in 1981 to a structuring presence by 1993.

Emergence of ecology. The Ecology and Animal Welfare topics appear almost exclusively in 1993, growing from near-zero to roughly 15% of the corpus combined. This coincides with the Entente des Écologistes (Verts + Génération Écologie) and marks the arrival of environmentalism as a structured electoral discourse rather than a fringe concern.

Decline of class-based rhetoric. The far-left themes (Workers' Struggles, Social Conflicts, Economic Sovereignty) are present in 1981 but shrink steadily through 1988 and 1993, consistent with the weakening of class-based political rhetoric and the fragmentation of far-left movements in the early 1990s. The Political Alternation topic is fading as the Left loses power.

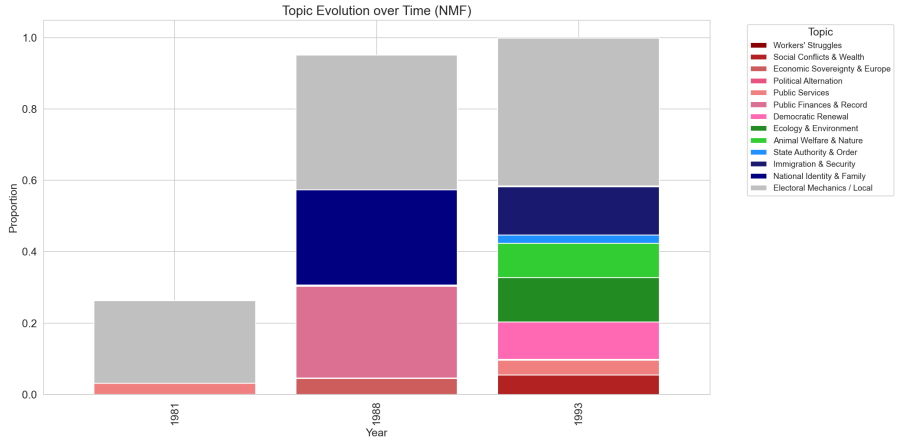


Figure 2: Evolution of NMF topic proportions across the 1981, 1988, and 1993 legislative elections.

6.3 Party Specialisation

To quantify thematic ownership beyond raw proportions, we compute the specialisation ratio $P(\text{topic} \mid \text{family})/P(\text{topic})$ for each family-topic couple. Values above 1.5 indicate significant over-representation; below 0.7, significant under-representation (Figure 3).

Extreme complementarity across the spectrum. The specialisation heatmap reveals an almost perfectly block-diagonal structure. The Far left dominates Workers' Struggles (9.28×), Social Conflicts (9.16×), Public Services (8.22×), and Economic Sovereignty (9.18×); all near zero for every other family. The Far right mirrors this pattern on Immigration (6.78×) and National Identity (6.81×). The Greens hold the highest single ratios in the dataset on Animal Welfare (8.80×) and Ecological Transition (8.46×). This complementarity suggests that French political discourse in this period was highly segmented, with minimal thematic overlap between the extremes.

Dynamic specialisation (1981 → 1993). The triple cross-tabulation (Figure 4) shows that these specialisations are not static. In 1981, the Far left over-indexes on Immigration (6.0×) reflecting an anti-capitalist framing of migration, or vouching for it, and not the identity-based framing that would come later and mentionned by the FN. Indeed, by 1993, Immigration has shifted entirely to the Far right (6.7×), while the Far left concentrates on Workers' Struggles (9.3×) and Social Conflicts (9.3×). The Left's profile also evolves: absent from Public Finances in 1981, it reaches 4.0× on that topic by 1993, consistent with the PS's shift from class-based rhetoric toward technocratic fiscal management during the second Mitterrand presidency. Several extreme ratios in 1988 (e.g. Greens at 252.6× on Animal Welfare) reflect the very small number of Green candidates that year, making these values statistically unstable. Overall, this dynamic view suggests that thematic ownership in French politics is not fixed but actively contested and transferred between families across electoral cycles. This finding would be invisible without the temporal disaggregation, these figures represent the output with the most upside for analysis.

Figure 3: NMF specialisation ratio $P(\text{topic} \mid \text{family})/P(\text{topic})$. Only significant values (ratio < 0.7 or > 1.5) are shown.

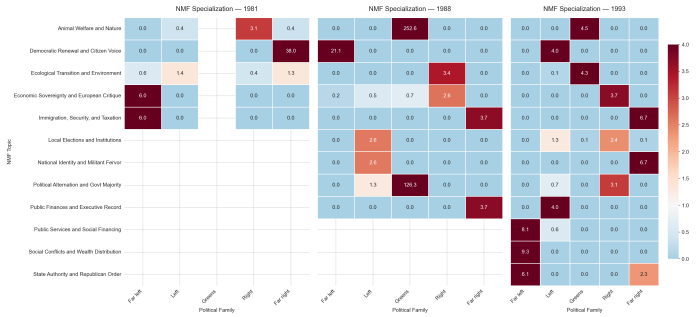


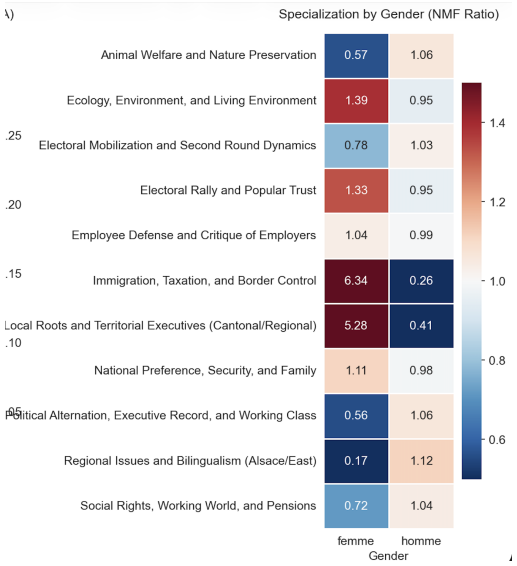
Figure 4: NMF specialisation ratio by election year. Empty cells indicate topics absent from that family-year combination.

6.4 Correlation Between Topics and Metadata

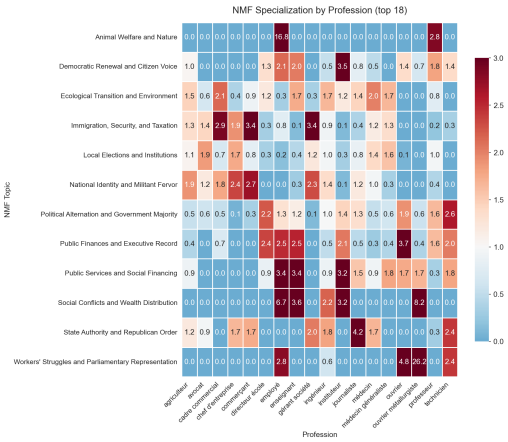
Gender. Cross-tabulating NMF topics with candidate gender shows that women over-index on Economic Sovereignty (5.87×), Animal Welfare (5.25×), and Workers' Struggles (3.15×), while men dominate the residual Local Elections category. However, women represented a small minority of candidates during this period, and their over-representation on far-left and green topics likely reflects the higher proportion of female candidates in those parties rather than a gender-specific thematic preference independent of affiliation. This confound limits the interpretability of the gender analysis.

Profession. Filtering to professions with at least 30 occurrences and computing the specialisation ratio provides a more robust sanity check. *Ouvriers* and *ouvriers métallurgistes* show extreme specialisation on Workers' Struggles (26.2×) and Social Conflicts (8.2×); *chefs d'entreprise* and *commerçants* lean toward Immigration/Security (3.0× and 3.4×); *enseignants* over-index on Public Services (3.4×). These alignments confirm that NMF topics capture genuine socio-professional variation : candidates from working-class backgrounds discuss class struggle, business owners

discuss security and taxation though the effect is partially mediated by the strong correlation between profession and party affiliation.



(a) Specialisation by gender (NMF)



(b) Specialisation by profession (NMF)

Figure 5: Correlation between NMF topics and candidate metadata. The profession heatmap is filtered to the 18 most frequent professions ($n \geq 30$).

7 Discussion and Conclusion

Our comparative evaluation demonstrates that NMF with TF-IDF weighting yields the most coherent and interpretable thematic structures for this corpus ($c_v = 0.66$), consistent with the literature on NMF's efficacy for short, focused documents. LDA serves as a robust baseline but frequently conflates substantive policy themes with generic electoral boilerplate. BERTopic successfully isolates specific phenomena but suffers from excessive granularity (195 topics) and a too high outlier rate (37%).

Beyond algorithmic performance, the cross-tabulation of topics with candidate metadata provides strong external validation. Partisan specialisation ratios align precisely with known ideological positions: the Far right concentrates on immigration and national identity, the Far left on workers' struggles and wealth redistribution, the Greens on ecology. The profession analysis confirms that *ouvriers* discuss class struggle and *chefs d'entreprise* discuss taxation, patterns that would be difficult to dismiss as artifacts. The volume of cross-analyses performed (by party, year, gender, profession, and all combinations) makes exhaustive interpretation impractical within this report; each heatmap contains dozens of data points worthy of discussion.

The most valuable insight arguably lies in the temporal dynamics of partisan discourse. The rise of immigration from a negligible theme in 1981 to a dominant one in 1993, the decline of class-based rhetoric on the Left, the emergence of ecology as a structured electoral discourse, and the shift of the Far left's framing of immigration from anti-capitalist to absent. This seems to fit with the Mitterand era. This suggests that topic modeling, despite its imperfections, is a powerful tool for large-scale political text analysis.

Limitations. Several caveats apply. First, 31% of documents lack political affiliation metadata and are excluded from partisan analyses at the very beginning, potentially introducing selection bias. Second, OCR quality varies across documents and election years, which may affect topic stability, for example the bilingual Alsace-Moselle tracts detected by LDA illustrate how corpus artifacts can surface as spurious topics. Third, manual labelling of both topics and political families involves subjective judgment. Finally, the residual Local Elections topic, which captures generic institutional language rather than a substantive theme, absorbs a large share of documents (particularly from the Right) and limits the depth of analysis for that family.

References

David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003.

David M. Blei. Probabilistic topic models. *Communications of the ACM*, 55(4):77–84, 2012.

Ricardo J.G.B. Campello, Davoud Moulavi, and Jörg Sander. Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD)*, pages 160–172, 2013.

Amy Catalinac. From pork to policy: The rise of programmatic campaigning in Japanese elections. *The Journal of Politics*, 78(1):1–18, 2016.

Rob Churchill and Lisa Singh. The evolution of topic modeling. *ACM Computing Surveys*, 54(10s):Article 215, 2022.

Derek Greene and James P. Cross. Exploring the political agenda of the European Parliament using a dynamic topic modeling approach. *Political Analysis*, 25(1):77–94, 2017.

Justin Grimmer and Brandon M. Stewart. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297, 2013.

Maarten Grootendorst. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*, 2022.

Daniel D. Lee and H. Sebastian Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791, 1999.

Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.

Kevin M. Quinn, Burt L. Monroe, Michael Colaresi, Michael H. Crespin, and Dragomir R. Radev. How to analyze political attention with minimal assumptions and costs. *American Journal of Political Science*, 54(1):209–228, 2010.

Michael Röder, Andreas Both, and Alexander Hinneburg. Exploring the space of topic coherence measures. In *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining (WSDM '15)*, pages 399–408, 2015.